

Multiple Choice Questions

1. Which of the following is NOT a characteristic of machine learning?

- a) Ability to learn from data
- b) Manual programming required
- c) Can handle large datasets
- d) Automation of complex tasks

Correct Answer: Manual programming required

Explanation: Machine learning algorithms are designed to learn from data without the need for explicit programming.

2. Which type of machine learning algorithm is used when the data is labeled?

- a) Supervised learning
- b) Unsupervised learning
- c) Semi-supervised learning
- d) Reinforcement learning

Correct Answer: Supervised learning

Explanation: Supervised learning algorithms use labeled data to learn the relationship between input features and output labels.

3. Which of the following is NOT a supervised learning algorithm?

- a) Linear regression
- b) Logistic regression
- c) Clustering

d) Support vector machines

Correct Answer: Clustering

Explanation: Clustering is an unsupervised learning algorithm used to group data points into clusters based on their similarities.

4. Which type of machine learning algorithm is used to reduce dimensionality in a dataset?

- a) Clustering
- b) Dimensionality reduction
- c) Anomaly detection
- d) Regression

Correct Answer: Dimensionality reduction

Explanation: Dimensionality reduction algorithms aim to reduce the number of features in a dataset while preserving important information.

5. Which of the following is a benefit of machine learning?

- a) Automation
- b) High accuracy
- c) Scalability
- d) All of the above

Correct Answer: All of the above

Explanation: Machine learning offers various benefits, including automation of tasks, high accuracy in predictions, and the ability to handle large datasets.

6. Which of the following is a challenge in machine learning?

- a) Data quality
- b) Overfitting
- c) Bias
- d) All of the above

Correct Answer: All of the above

Explanation: Machine learning models can be affected by data quality, overfitting, and biases inherent in the data.

7. Which of the following is an application of machine learning?

- a) Product recommendations
- b) Spam filtering
- c) Healthcare diagnosis
- d) All of the above

Correct Answer: All of the above

Explanation: Machine learning has various applications, including product recommendations, spam filtering, and healthcare diagnosis.

8. What is the main goal of unsupervised learning algorithms?

- a) To map input features to output labels
- b) To identify patterns and structures in data
- c) To predict continuous numerical values
- d) To classify data into different categories

Correct Answer: To identify patterns and structures in data

Explanation: Unsupervised learning algorithms aim to find patterns or structures in unlabeled data

without any guidance.

9. Which of the following is NOT a common supervised learning algorithm?

- a) Linear regression
- b) Logistic regression
- c) Nave Bayes
- d) Clustering

Correct Answer: Clustering

Explanation: Clustering is an unsupervised learning algorithm, while the others are supervised learning algorithms.

10. How does data quality affect machine learning models?

- a) Can lead to inaccurate or unreliable results
- b) Does not have a significant impact on model performance
- c) Improves model performance even if the data is noisy
- d) Makes models more robust to overfitting

Correct Answer: Can lead to inaccurate or unreliable results

Explanation: Poor data quality can hinder the ability of machine learning models to learn effectively and make accurate predictions.